

Multi-Objective Pareto Hyperparameter Optimization for Joint Detection and Localization of Forged Identity Documents

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Abstract

Automated forensic analysis of identity documents requires simultaneously detecting whether a document has been tampered with and identifying which regions were altered. These two objectives are structurally coupled but exist in tension: optimizing one with a fixed loss weight may degrade the other. We propose DocVerify, a multi-task convolutional network paired with a multi-objective hyperparameter optimization (HPO) framework that treats detection quality (PR-AUC) and localization quality (Dice) as simultaneous objectives. Using the Multiobjective Tree-structured Parzen Estimator (MOTPE) within a 10-fold nested cross-validation protocol (50 trials per outer fold, 500 unique configurations evaluated across 5 inner splits — 2 500 individual model trainings in total), DocVerify selects configurations from the Pareto-optimal front by minimizing Euclidean distance to the ideal point (1, 1), without any manual weighting. Evaluated on the FantasyID benchmark (Korshunov et al., 2025a), DocVerify achieves PR-AUC = 0.9921 ± 0.0058 and Dice = 0.856 ± 0.030 under nested cross-validation, and detection F1@0.5 = 0.969 ± 0.014 with localization F1 = 0.807 ± 0.096 on the blind test set (30 seeds). An ablation confirms that Pareto-optimized loss weighting provides a statistically significant Dice improvement over equal-weight joint training ($p = 1.3 \times 10^{-4}$, $d = 1.01$, large effect) while maintaining equivalent detection accuracy.

Keywords: identity document forensics, forgery detection, forgery localization, multi-task learning, multi-objective HPO, Pareto front, nested cross-validation, MOTPE

1. Introduction

Identity document fraud poses a growing threat to national security, financial integrity, and border control worldwide. The widespread availability of image-editing software has made it possible to produce convincing forgeries of passports, national identity cards, and driver’s licenses at low cost (Europol Innovation Lab, 2022). Automated forensic analysis of identity documents is therefore a critical component of modern digital identity verification pipelines.

Detection of forged identity documents is inherently a two-level problem. At the *document level*, a system must decide whether a presented document is genuine or has been tampered with (binary detection). At the *pixel level*, it must also identify and delineate which specific regions have been altered (forgery localization). Both objectives are operationally necessary.

Binary detection alone is insufficient for real-world forensic deployments. A positive detection decision provides no *actionable information* to the investigator: it does not reveal whether the fraudster altered the portrait photograph, falsified textual fields, or combined multiple manipulations. Without spatial evidence, a human expert cannot determine the nature of the fraud or build a legal case. Detection systems are also increasingly subject to *explainability requirements* under emerging AI regulations (European Parliament and Council of the European

Union, 2024): a binary “tampered” label cannot satisfy the obligation to provide a meaningful explanation of the automated decision. Conversely, localization alone does not yield a reliable document-level confidence score suitable for automated accept/reject decisions. A practical forensic system must address both tasks simultaneously.

The primary technical difficulty in joint detection and localization is that the two objectives are in *structural tension*: the classification head benefits from global image features, while the segmentation head requires spatially precise, pixel-level representations. Prior work either treats the two tasks independently (Guillaro et al., 2022) or collapses both losses into a scalar objective via a fixed weight λ (Caruana, 1997; Cipolla et al., 2018). The choice of λ is arbitrary and dataset-dependent. A second challenge is the risk of over-fitting the model-selection criterion: HPO that maximizes PR-AUC may sacrifice Dice, and vice versa.

This paper introduces DocVerify, a multi-task forensic document analysis system that addresses both challenges. Our contributions are:

1. *Multi-objective Pareto HPO for multitask forensics.* Both PR-AUC and Dice are treated as simultaneous optimization objectives. The optimal configuration is selected from the Pareto front by minimizing the Euclidean distance to the ideal point (1, 1), requiring no manual task-weight tuning.
2. *DocVerify architecture.* An end-to-end multi-task CNN comprising a lightweight convolutional encoder coupled

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to a U-Net decoder for pixel-level mask prediction and a classification head for document-level detection, trained jointly with a combined BCE+Dice segmentation loss.

3. *Rigorous nested cross-validation.* A 10-fold outer $\times$ 5-fold inner nested cross-validation scheme with 50 MOTPE trials per outer fold (2 500 HPO evaluations total). Final performance is estimated on a blind holdout using 30 independent seeds, with Wilcoxon signed-rank tests and Holm–Bonferroni correction.

The source code and experimental artifacts are publicly available at <https://github.com/HernanDiaz/IDVerify/>.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the DocVerify architecture and HPO framework. Section 4 presents experimental results. Section 5 discusses the findings and limitations. Section 6 concludes.

2. Related Work

2.1. Document and Image Forgery Detection

Deep learning has substantially advanced general-purpose image forensics: CNNs can detect subtle statistical fingerprints left by editing operations, often outperforming hand-crafted pipelines (Bappy et al., 2019; Wu et al., 2019). Cozzolino and Verdoliva (2020) (Noiseprint) train a CNN via Siamese contrastive learning to extract camera-model noise residuals; forgeries appear as anomalies in this residual without requiring semantic understanding. Guillard et al. (2022) (TruFor) generalize this with a transformer fusion network (NoisePrint++) that jointly models RGB semantics and noise-pattern anomalies, achieving strong performance on general image forensics benchmarks with over 876 000 pre-training images.

Identity document forgery introduces unique challenges: forged regions are often small (a single text field or portrait), the structured background (guilloché patterns, UV-reactive inks) creates strong false negatives for general-purpose noise detectors, and document type diversity demands generalization beyond natural-image pre-training. Official evaluations on FantasyID confirm that general-purpose baselines (TruFor, MMFusion, FatFormer, UniFD) all achieve false-negative rates above 50% at a 10% false-positive threshold (Korshunov et al., 2025a), motivating domain-specific approaches.

Among systems trained exclusively on FantasyID, George and Marcel (2025) propose EdgeDoc, a lightweight CNN-Transformer encoder (EdgeNeXt) fed with the green channel and a NoisePrint residual, with a U-Net decoder optimized jointly for classification and localization. Yu et al. (2025) introduce Re-MTKD, a reinforcement-learning multi-teacher knowledge distillation framework using a ConvNeXt v2 encoder; the Sunlight team adapted this to identity documents using over 60 000 additional external images, winning both tracks of the DeepID 2025 Challenge (Korshunov et al., 2025b).

DocVerify differs from all prior document-forensics systems in three respects: (i) it systematically searches the task-weight λ_{mask} via multi-objective HPO rather than fixing it heuristically; (ii) it selects the final configuration via a Pareto-optimal criterion;

and (iii) it reports uncertainty through nested cross-validation, providing statistically rigorous performance estimates.

2.2. Multi-Task Learning and HPO

In multi-task learning (Caruana, 1997; Ruder, 2017), the dominant approach combines task losses via scalar scalarization (Marler and Arora, 2004) with fixed weights set before training. Cipolla et al. (2018) propose learning these weights via homoscedastic uncertainty; Liu et al. (2019) introduce gradient normalization. All scalar approaches commit to a fixed trade-off before training begins.

Multi-objective HPO extends single-objective search to jointly optimize multiple metrics (Miettinen, 1999). The Multiobjective TPE (MOTPE) (Ozaki et al., 2022) adapts the TPE surrogate (Bergstra et al., 2011) to the multi-objective setting, achieving competitive Pareto-front coverage with far fewer function evaluations than evolutionary methods such as NSGA-II (Deb et al., 2002)—critical when each evaluation requires training a deep network. To our knowledge, this is the first work to apply Pareto-based multi-objective HPO to joint identity-document forgery detection and localization within a nested cross-validation framework.

3. Methodology

3.1. Dataset

We train and evaluate DocVerify on FantasyID (Korshunov et al., 2025a), a publicly released benchmark for identity document forgery detection and localization developed by Idiap Research Institute for the DeepID 2025 Challenge (Korshunov et al., 2025b). FantasyID comprises approximately 3 284 labeled images of synthetic fantasy identity documents: *bonafide* (genuine) and *attack* (tampered via face swap or text inpainting).

Each image is accompanied by JSON annotations specifying bounding-box regions with `region_provenance`: `altered`. We convert these to binary segmentation masks at training resolution (224×224 pixels). We apply GroupShuffleSplit to all 3 284 available FantasyID images (grouping by document identity, `test_size` = 0.15, `random_state`=42) to obtain approximately 2 791 development images and 493 blind holdout images, ensuring all captures of the same document identity remain in the same partition. The dataset is class-imbalanced: 2 351 attack images (71.6%) and 933 bonafide images (28.4%); this imbalance is preserved in all splits and not corrected via resampling or loss weighting.

3.2. Architecture

DocVerify is a multi-task encoder-decoder network for simultaneous document-level classification and pixel-level forgery localization (Fig. 1).

PATEL CNN ENCODER

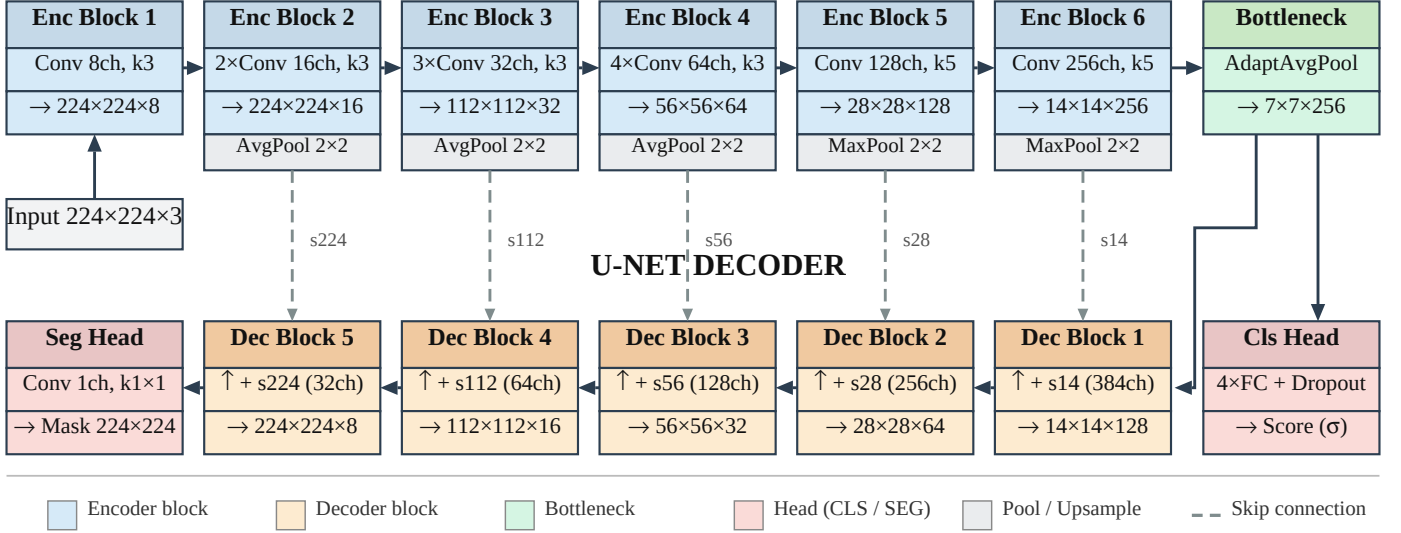


Figure 1: DocVerify architecture. The encoder maps a $3 \times 224 \times 224$ input to a $256 \times 7 \times 7$ bottleneck through six convolutional blocks ($8 \rightarrow 256$ channels). The U-Net decoder reconstructs the forgery mask via skip connections; the classification head outputs a document-level tampered/genuine score.

3.2.1. Encoder

The encoder is a six-block lightweight CNN inspired by [Patel et al. \(2023\)](#), a deepfake-detection architecture specifically designed for forgery artifact extraction. A lightweight encoder is a deliberate design choice: the nested CV protocol requires training ~ 2500 models, so per-trial cost must be kept low (~ 3.5 min on our hardware). Heavier pretrained backbones (e.g. ResNet-50, ViT) would make this scale infeasible; moreover, with only ~ 2800 training images, large pretrained encoders risk importing ImageNet biases that are not necessarily beneficial for structured-document forgery detection. The channel progression is $8 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ filters. All layers use LeakyReLU ($\alpha = 0.2$) ([Xu et al., 2015](#)) and batch normalization ([Ioffe and Szegedy, 2015](#)). Blocks 1–4 use 3×3 convolutions; blocks 5–6 use 5×5 convolutions. Downsampling uses average pooling after blocks 2 and 3 (to preserve boundary information) and max pooling after blocks 4 and 5 (for translation invariance). Skip connections are extracted from each block prior to pooling.

3.2.2. Classification Head

Global average pooling is applied to the 7×7 bottleneck, yielding a 256-dimensional vector passed through four fully-connected layers ($256 \rightarrow 32 \rightarrow 16 \rightarrow 16 \rightarrow 1$) with dropout and LeakyReLU, producing a classification logit $\hat{y}_{\text{cls}}$.

3.2.3. Segmentation Head (U-Net Decoder)

A five-stage U-Net decoder ([Ronneberger et al., 2015](#)) reconstructs the spatial mask via bilinear upsampling and skip-connection concatenation. The channel width halves at each stage from C_{dec} (a tunable HPO parameter) down to $C_{\text{dec}}/16$. A final 1×1 convolution produces the segmentation logit $\hat{Y}_{\text{mask}}$.

3.3. Loss Functions

The combined training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{mask}} \mathcal{L}_{\text{seg}}, \quad (1)$$

where $\lambda_{\text{mask}} \in [0.5, 3.0]$ is included in the HPO search space. $\mathcal{L}_{\text{cls}} = \text{BCEWithLogits}(\hat{y}_{\text{cls}}, y_{\text{cls}})$. The segmentation loss combines BCE and Dice:

$$\mathcal{L}_{\text{seg}} = \text{BCE}(\hat{Y}, Y) + 1 - \frac{2 \sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}, \quad (2)$$

as introduced for volumetric segmentation ([Milletari et al., 2016](#)) to address foreground/background imbalance in small forgery regions.

All models are trained with AdamW ([Loshchilov and Hutter, 2019](#)) and ReduceLROnPlateau (factor 0.5, patience 3). Mixed-precision training (FP16/FP32 via PyTorch AMP ([Micikevicius et al., 2018](#))) is used throughout.

3.4. Multi-Objective HPO

We optimize two complementary metrics: *PR-AUC* (area under the precision-recall curve, measuring document-level detection quality) and *Dice* ($2|P \cap G|/(|P| + |G|)$, measuring pixel-level mask overlap). Our HPO formulation simultaneously maximizes:

$$\max_{\theta \in \Theta} (\text{PR-AUC}(\theta), \text{Dice}(\theta)). \quad (3)$$

Table 1 defines the search space Θ .

We use MOTPE ([Ozaki et al., 2022](#)) as implemented in Optuna ([Akiba et al., 2019](#)). After 50 trials on the inner splits, we select the configuration θ^* from the Pareto front $\mathcal{P}$ by minimizing the Euclidean distance to the ideal point:

$$\theta^* = \arg \min_{\theta \in \mathcal{P}} \sqrt{(1 - \text{PR-AUC})^2 + (1 - \text{Dice})^2}. \quad (4)$$

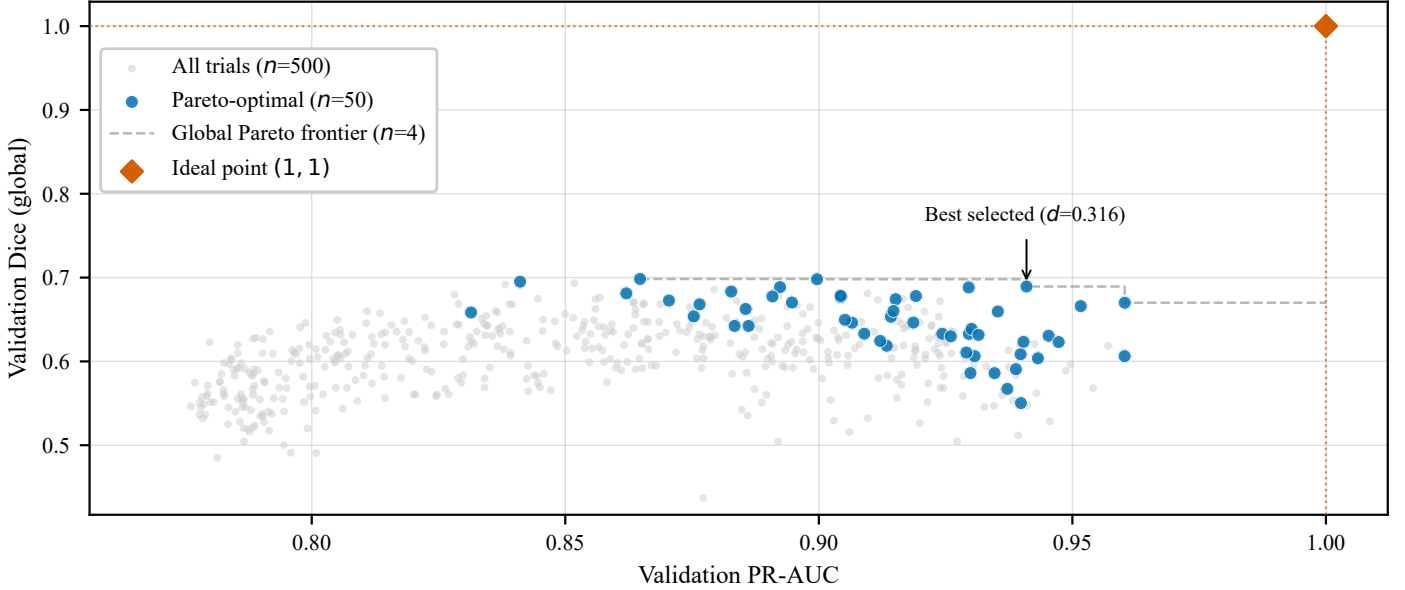


Figure 2: Validation objective space across all 500 HPO trials (10 folds $\times$ 50 MOTPE trials). Grey: evaluated; blue: Pareto-optimal; dashed: Pareto frontier. Red diamond: ideal point (1, 1); annotated point: selected configuration (distance $d = 0.316$).

Table 1: HPO search space.

Parameter	Distribution	Range
η (learning rate)	log-uniform	$[5 \times 10^{-5}, 9 \times 10^{-4}]$
λ_{wd} (weight decay)	log-uniform	$[10^{-7}, 10^{-4}]$
p (dropout)	uniform	$[0.1, 0.4]$
C_{dec} (decoder channels)	categorical	{96, 128, 192, 256}
λ_{mask} (mask loss weight)	uniform	$[0.5, 3.0]$

This criterion requires no user-specified weighting and assigns equal importance to detection and localization. Figure 2 illustrates the resulting objective space.

3.5. Nested Cross-Validation

To obtain unbiased estimates while keeping HPO blind to test data, we use a three-level nested CV scheme (Varma and Simon, 2006). The outer loop uses 10 stratified folds; each fold serves once as the test set. For each outer fold, the remaining data is split into 5 inner folds on which 50 MOTPE trials are run (up to 15 epochs each, patience = 1). The optimal θ^* is then used to retrain from scratch on all outer-fold training data (up to 100 epochs, early stopping patience = 12). This design ensures HPO never sees test data: 50 MOTPE trials per outer fold yield 500 unique hyperparameter configurations in total (10 outer folds $\times$ 50 trials); because each configuration is evaluated on 5 inner splits, this corresponds to $10 \times 50 \times 5 = 2\,500$ individual model trainings, plus 10 final models retrained on full outer-fold data.

4. Experiments and Results

4.1. Experimental Setup

All experiments are conducted on a single workstation running Windows 11, equipped with an AMD Ryzen 5

3400G (3.70 GHz), 32 GB DDR4-3200 RAM, and an NVIDIA RTX 5060 Ti (16 GB VRAM), using PyTorch 2.10.0 with CUDA 13.0 and Optuna 3.x. Training uses a batch size of 64 and no data augmentation: all images are resized to 224×224 pixels and normalised to $[0, 1]$ prior to caching in GPU memory. The full NCV pipeline required approximately 59 hours; VRAM caching reduces per-trial training time from ~ 12 to ~ 3.5 minutes.

4.2. Nested Cross-Validation Results

Table 2 reports per-fold test metrics. PR-AUC is consistently high across all folds (0.977–0.999, mean 0.9921 ± 0.0058). Dice shows slightly more fold-to-fold variability (0.800–0.909, mean 0.856 ± 0.030), reflecting the difficulty of pixel-level localization. The Pareto-selected λ_{mask} clusters between 2.3 and 2.7 in eight of ten folds; the two outlier folds (F5, F9) correspond to the two lowest Dice scores, confirming that λ_{mask} is a decisive driver of localization quality.

4.3. Ablation Study

To isolate each design decision, we evaluate four variants on the blind test set using 30 independent seeds: *Multitask* (ours, joint detection + localization with Pareto-optimized λ_{mask}); *cls_only* (classification head only); *seg_only* (segmentation head only, detection by thresholding mean mask probability); and *unweighted_losses* (full multitask, $\lambda_{cls} = \lambda_{mask} = 0.5$, bypassing HPO). Table 3 summarizes results.

Three findings stand out. First, joint multitask training is essential: *cls_only* produces near-zero Dice (0.002 ± 0.008). Second, a dedicated classification head is necessary: *seg_only* achieves competitive Dice (0.867) but PR-AUC drops to 0.728 ± 0.053 , confirming that the segmentation logit alone is an unreliable detection signal. Third, Pareto-optimized weights outperform equal weights specifically on localization:

Table 2: Per-fold test results from 10-fold NCV. $\bar{w}_{\text{mask}}$: Pareto-selected loss weight.

Fold	PR-AUC	Dice	$\bar{w}_{\text{mask}}$
F1	0.9939	0.861	2.46
F2	0.9769	0.871	2.70
F3	0.9993	0.909	2.47
F4	0.9915	0.858	2.43
F5	0.9948	0.827	1.41
F6	0.9936	0.874	2.89
F7	0.9940	0.861	2.67
F8	0.9967	0.887	2.29
F9	0.9896	0.807	1.95
F10	0.9909	0.800	2.65
Mean	0.9921	0.856	2.39
Std	0.0058	0.030	0.43

`unweighted_losses` matches Multitask on PR-AUC ($p = 0.083$) but shows significantly lower Dice ($p = 1.3 \times 10^{-4}$, $d = 1.01$, large effect), validating the HPO contribution.

4.4. Pareto HPO vs. Scalar Selection

A key question for any multi-objective HPO framework is whether the added complexity over simpler scalar alternatives yields measurable gains. We address this directly by comparing three selection criteria applied to the *same* 50 MOTPE trials per outer fold: Pareto (ours, distance to ideal point); `by_prauc` (select trial maximizing inner-fold PR-AUC); and `by_dice` (select trial maximizing inner-fold Dice). Results on the 10-fold NCV are shown in Table 4.

No statistically significant differences are found between any pair of selection criteria (Wilcoxon + Holm, all $p \geq 0.05$, $n = 10$), with absolute differences below 0.004 on PR-AUC and 0.008 on Dice. This result should be read as a *positive* finding: Pareto selection matches scalar alternatives at no cost in predictive accuracy, while delivering methodological benefits that scalar criteria cannot provide. It requires no a priori commitment to a primary objective, makes the full trade-off landscape observable (Fig. 2), and is structurally consistent with the DeepID challenge protocol, which ranks detection and localization independently.

4.5. Comparison with Related Systems

Table 5 compares DocVerify against the top participants of the DeepID 2025 Challenge on Track 1 (detection F1@0.5) and Track 2 (per-image localization F1), computed on our 15% holdout using 30 seeds.

This comparison must be interpreted with caution. DocVerify was developed independently after the DeepID 2025 Challenge submission deadline had already closed; official test-set evaluation was therefore not possible and results are reported on our internal 15% holdout, whereas all challenge figures were obtained on the official test set (FantasyID test partition plus a

private out-of-domain set from PXL Vision). The two evaluation sets are not identical, making direct numerical comparisons indicative rather than strictly controlled.

It is also important to note the training data disparity among the systems in Table 5. Sunlight augmented the FantasyID training set with over 60 000 external identity document images; the TruFor baseline was pre-trained on 876 000 general-purpose manipulation images before any fine-tuning. By contrast, AG/EdgeDoc, UAM-Biometrics, and DocVerify trained exclusively on FantasyID, making them more directly comparable to one another.

With that caveat, DocVerify achieves $F1@0.5 = 0.969 \pm 0.014$ on Track 1, broadly in line with the other FantasyID-only systems (AG/EdgeDoc: 0.958; UAM-Biometrics: 0.712) and competitive with Sunlight (0.991) despite using no external data. On Track 2, DocVerify achieves $F1_{\text{loc}} = 0.807 \pm 0.096$, comparable to the challenge winner Sunlight (0.784); however, the margin (+0.023) is well within the standard deviation of our estimates and should not be interpreted as a definitive improvement. The high Track 2 variance is driven by seeds where bonafide misclassifications generate false-positive mask pixels; F1 on confirmed attack images is substantially more stable (0.881 ± 0.015).

4.6. Qualitative Analysis

Figure 3 shows a representative output for a forged document with five altered regions. The predicted mask successfully localizes all five tampered fields — photograph, given name, date of birth, document number, and expiry date — with tight spatial alignment to the ground-truth alterations.

5. Discussion

5.1. Analysis of Key Findings

The ablation reveals an asymmetry between the two tasks. The `seg_only` variant achieves competitive Dice (0.867 ± 0.023 , $p = 0.288$ vs. Multitask) but its PR-AUC collapses to 0.728 ± 0.053 : the U-Net decoder captures sufficient forgery evidence for pixel-level prediction yet cannot reliably aggregate it into a binary document-level decision. The result confirms that the shared encoder, regularized by the segmentation loss, learns spatially-aware features that strongly improve document-level detection.

Comparing Multitask against `unweighted_losses` isolates the contribution of HPO loss weighting. These variants differ only in whether λ_{mask} is set by Pareto HPO (mean 2.39 ± 0.43) or fixed at 0.5. Pareto-optimized weights yield significantly higher Dice ($p = 1.3 \times 10^{-4}$, $d = 1.01$, large effect) while maintaining equivalent PR-AUC, confirming that $\lambda_{\text{mask}} \approx 2.4$ is broadly optimal and that MOTPE surfaces this value reliably across folds.

5.2. Limitations

Single dataset. All results are obtained on FantasyID, a synthetic dataset from a single research institute. Generalization to real-world identity documents, unseen document designs, or physical tampering attacks remains an open question.

Table 3: Ablation study (blind test, 30 seeds). p : Wilcoxon + Holm; d : Cohen’s d . *** $p < 0.001$; ns $p \geq 0.05$.

Variant	PR-AUC		Dice			
	Mean $\pm$ Std	vs. Multitask	Mean $\pm$ Std	vs. Multitask	p_{holm}	d
Multitask (ours)	0.9967 $\pm$ 0.0021	—	0.875 $\pm$ 0.018	—	—	—
cls_only	0.9503 $\pm$ 0.0744	***	0.002 $\pm$ 0.008	***	1.3×10^{-8}	large
seg_only	0.7277 $\pm$ 0.0528	***	0.867 $\pm$ 0.023	ns	0.288	small
unweighted_losses	0.9958 $\pm$ 0.0016	ns	0.857 $\pm$ 0.016	***	1.3×10^{-4}	1.01

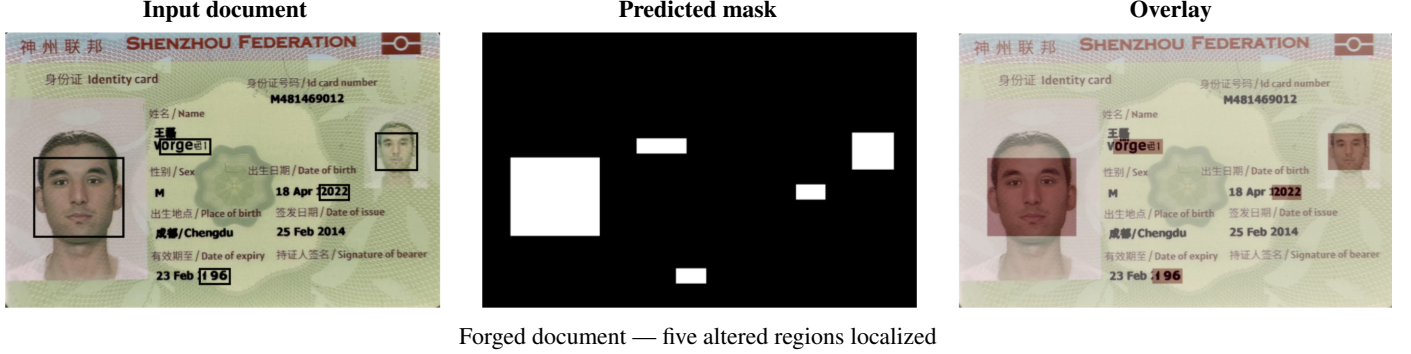
Figure 3: Qualitative result on a forged FantasyID sample. **Left:** input document. **Centre:** binary mask predicted by DocVerify. **Right:** predicted mask overlaid in red on the input. All five tampered fields are correctly localized.

Table 4: Pareto vs. scalar selection on 10-fold NCV (same 50 MOTPE trials).

Method	PR-AUC	Dice
Pareto (ours)	0.9921 $\pm$ 0.0060	0.856 $\pm$ 0.035
by_prauc	0.9958 $\pm$ 0.0015	0.857 $\pm$ 0.017
by_dice	0.9936 $\pm$ 0.0033	0.863 $\pm$ 0.015

Table 5: Comparison with DeepID 2025 Challenge systems on Track 1 (detection F1@0.5) and Track 2 (per-image localization F1). †: official challenge test set; DocVerify on internal holdout (see text). Dashes: result not reported.

System	F1@0.5	F1 _{loc}
Sunlight (Yu et al., 2025) (1st)†	0.991	0.784
Incode (2nd)†	0.868	—
AG/EdgeDoc (George and Marcel, 2025) (3rd)†	0.958	0.621
UAM-Biometrics (4th)†	0.712	—
TruFor (Guillaro et al., 2022)	0.807	0.590
baseline†	—	—
DocVerify (ours)	0.969 $\pm$ 0.014	0.807 $\pm$ 0.096

Architecture. The lightweight encoder was chosen to make 2500-model nested CV tractable; richer pretrained backbones (Dosovitskiy et al., 2021) are expected to improve cross-domain generalization but would preclude the rigorous HPO validation that is the paper’s primary contribution.

Localization variance. The high Track 2 standard deviation (± 0.096), driven by bonafide false-positive masks, represents a practical reliability concern. Targeted calibration strategies are a

natural extension.

6. Conclusion

We presented DocVerify, a multi-task CNN for joint detection and localization of forged identity documents, paired with a multi-objective Pareto HPO framework that selects hyperparameter configurations by optimizing PR-AUC and Dice simultaneously without fixing their relative importance in advance.

Our main findings are: (i) joint multitask training with Pareto-optimized loss weights achieves significantly higher Dice than single-task variants and equal-weight joint training (blind test, 30 seeds, $p = 1.3 \times 10^{-4}$, $d = 1.01$) while maintaining competitive PR-AUC; (ii) under a rigorous 10-fold nested cross-validation protocol, DocVerify achieves PR-AUC = 0.9921 $\pm$ 0.0058 and Dice = 0.856 $\pm$ 0.030, demonstrating consistent generalization across data partitions; and (iii) training exclusively on FantasyID without any external pre-training, DocVerify achieves results broadly competitive with top DeepID 2025 Challenge participants (F1@0.5 = 0.969 $\pm$ 0.014; F1_{loc} = 0.807 $\pm$ 0.096 on our internal holdout), though direct comparison is indicative given that challenge systems were evaluated on a different official test set.

Future work includes cross-domain evaluation on additional document datasets, integration of forgery-artifact-centric features (noise fingerprints, frequency anomalies (Cozzolino and Verdoliva, 2020)) for domain-invariant representations, and foundation model backbones (Muñoz-Haro et al., 2025) for richer pre-training.

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